Intellectual Property Risks in AI

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 25: AI Risk Management

1. Why AI IP Risk Is Different From Traditional IP Risk

Intellectual property risk in AI is different from traditional software IP risk in scale, mechanism, and legal ambiguity. Traditional software copyright protects specific code expression; AI systems trained on copyrighted data transform that data in ways that may or may not constitute infringement depending on contested legal theories. Traditional patent risk involves specific algorithm implementations; AI patent risk involves foundational techniques — attention mechanisms, transformer architectures, optimization objectives — that underpin entire industry categories. Traditional trade secret protection covers source code and formulas; AI trade secret protection must cover model weights, training datasets, and training methodologies, all of which can be threatened by model extraction attacks that require no internal access. The litigation landscape changed dramatically between 2022 and 2025: artists sued Stability AI and Midjourney, the New York Times sued OpenAI and Microsoft, GitHub Copilot faced class action. These are active proceedings that will define operating rules for AI IP for a decade. For the SecAI+ exam, IP risk is a first-class governance concern — ISO/IEC 42001:2023 Section 6.1 requires IP risks to be part of the AI management system's formal risk assessment.

2. Training Data Copyright: The Active Legal Battleground

The central unresolved legal question in AI IP is whether using copyrighted content to train a model infringes the copyright holder's exclusive reproduction and distribution rights. AI companies argue that training is transformative use protected by fair use doctrine. Copyright holders argue that ingesting their work into a training pipeline without license or compensation is unauthorized reproduction. The New York Times lawsuit against OpenAI (filed December 2023) included examples of the model reproducing Times articles nearly verbatim — a critical argument because memorization weakens the transformative use defense. In the EU, the Digital Single Market Directive's text and data mining exception allows training without permission for non-commercial research, but commercial AI training faces tighter constraints. The practical implication: if you are training on external data, you must know what you are training on. Assuming fair use applies without legal analysis is not a defensible governance posture.

3. US Fair Use Analysis for AI Training Data

Fair Use Factor	What Courts Consider	AI Training Implication
1. Purpose and character	Is the use transformative? Commercial or non-commercial?	AI companies argue transformation; courts have not definitively agreed
2. Nature of copyrighted work	Factual works receive less protection than creative works	Images, fiction, and journalism receive stronger protection than factual databases
3. Amount and substantiality	How much was taken, and was it the 'heart' of the work?	Training on complete works at scale is harder to justify than using excerpts
4. Market effect	Does the use harm the market	AI-generated content in the same genre as

	for the original?	training data may constitute direct market harm

Factors 1 and 4 are doing the most analytical work in active AI copyright litigation. No definitive US federal court ruling has resolved training data fair use as of 2025. The exam point: do not assume fair use applies. Legal counsel with AI IP specialization is required for training data decisions involving copyrighted content.

4. Data Provenance Management: The Practical Response

The operational response to training data copyright risk is systematic data provenance management: knowing exactly what data you trained on, where it came from, and what license governs its use. Training data with clear licensing provides a dramatically stronger IP position than data scraped from sources with no explicit license.

- **Common Crawl:** widely used in LLM pretraining but contains substantial copyrighted content with no explicit license for commercial training use — significant IP exposure.
- **Licensed datasets:** RedPajama, OpenWebText, and curated versions of The Pile document licensing and attempt to exclude restricted content — stronger IP position.
- **Proprietary licensed datasets:** content licensed directly from publishers, news organizations, or data APIs with explicit commercial training rights — the clearest IP position but highest cost.
- **Data lineage tools:** tools that track which data was used to produce which model version, creating the audit trail needed to respond to IP claims. Without provenance documentation, you cannot prove what you trained on even if you wanted to demonstrate compliance.

5. Real-World Incident: Getty Images v. Stability AI

Real-World Incident — Getty Images v. Stability AI (2023): Getty Images filed suit against Stability AI in both US and UK courts, alleging that Stable Diffusion was trained on over 12 million Getty-licensed images without authorization. The lawsuit included examples of generated images reproducing Getty watermarks — a striking piece of evidence suggesting memorization. The case demonstrates that creative industry rights holders have both the motivation and the evidence to pursue training data copyright claims. For AI developers, it illustrates the specific risk of training on watermarked or attributed content where memorization can be demonstrated forensically, and the reputational damage that attaches to copyright litigation alongside legal liability. Data provenance documentation and licensing agreements are the risk controls the case argues for.

6. Model Ownership and IP Assignment

When an organization develops an AI model, ownership of that model is a balance-sheet asset — but only if the right contractual agreements are in place. Common failure modes include:

- Employment agreements without explicit 'work for hire' language — absent this, the common law presumption in some jurisdictions may not assign all IP to the employer.

- Contractor or vendor agreements without IP assignment clauses — the contractor may retain rights to the work product, including model components they developed.
- Collaborative development across institutions without a pre-negotiated IP ownership agreement — IP ownership becomes a dispute after the model is complete.
- Research partnerships where academic institution co-inventors retain rights under Bayh-Dole or equivalent frameworks, creating co-ownership complications for commercial exploitation.

The model itself — its weights, architecture, and training methodology — may be protectable as trade secret (if kept confidential) and potentially patent (if the training process is novel and non-obvious). Get the legal agreements in place before development begins, not after the asset exists.

7. Trade Secrets: Protecting Your AI Assets

If competitive advantage comes from a proprietary model, unique training dataset, or novel training methodology, trade secret law may provide protection without patent disclosure requirements. A trade secret is confidential business information that provides a competitive advantage and is subject to reasonable efforts to maintain its secrecy. For AI, protectable trade secrets include model weights, curated training datasets, fine-tuning techniques, and architecture decisions. Maintaining trade secret status requires:

Requirement	Implementation
Enforceable access controls	Role-based access limiting model weights and training data to authorized personnel; audit logging of all access
NDAs with employees and contractors	Explicit NDA coverage for AI-specific confidential information, not just generic IP clauses
Compartmentalization	Ensure no single person has access to all components — training data expert, architecture expert, and inference optimization expert are separate roles
Audit trails	Demonstrating who accessed what and when is essential for pursuing misappropriation claims

The 2024 criminal charges against a former Google engineer for allegedly stealing AI trade secrets for the benefit of Chinese AI companies illustrates that AI model theft is prosecuted as a serious criminal offense. The case also illustrates the insider threat vector — the individual with legitimate access is the primary risk.

> **Compartmentalization as Trade Secret Defense:** No single departing employee should be able to take the complete IP picture. The person who knows the training data does not have access to the architecture details, and vice versa. This is both a trade secret maintenance requirement and an insider threat control.

8. Model Extraction Attacks: Trade Secret Theft via API

Model extraction attacks — querying a production model extensively and using input-output pairs to train a substitute model — constitute a form of trade secret theft when the target model represents proprietary IP. A 2020 paper by Tramer et al. demonstrated that models could be extracted with high fidelity through API queries alone, without access to original weights or training data. The extracted model can compete directly with the original, using the owner's R&D investment without authorization. MITRE ATLAS v2.1 classifies this under AML.T0036 (ML Model Inference API Access) combined with exfiltration techniques. Defenses include:

- Query rate limiting — restrict the number of queries per account/IP per time period to prevent systematic probing.
- Anomaly detection for systematic query patterns — flag accounts running grid searches across the input space.
- **Output watermarking:** embedding imperceptible signals in model outputs that allow detection of models trained on those outputs — enables forensic identification of extracted models.
- Prediction perturbation — adding calibrated noise to outputs that degrades extraction fidelity without significantly affecting legitimate use.

9. Open-Source AI Licensing: Not All Open Is Equal

The open-source AI ecosystem expanded rapidly after 2022, but licensing terms vary enormously and commercial use requires careful review. Key frameworks to know for the SecAI+ exam:

License / Model	Commercial Use	Key Restriction
Apache 2.0 (Mistral, Falcon)	Permissive commercial use	Attribution required; patent grant included
Meta LLaMA 2 Custom License	Permitted for orgs < 700M monthly active users	Cannot use outputs to train competing foundation models; acceptable use policy must be agreed
RAIL (Responsible AI License)	Use-case restricted	Behavioral restrictions prohibit specific harmful applications even for permissive licensees
Custom research-only licenses	Research only	Commercial deployment of any kind is prohibited

Before integrating any open model into a commercial product, read the full license text and confirm with legal counsel that the specific use case is permitted. API providers change their terms — build monitoring for TOS changes into vendor management processes.

10. Output Ownership and Third-Party API Terms

When a user generates content using an AI system — a generated image, co-authored document, or AI-assisted code — who owns that output? The US Copyright Office has consistently held that works created purely by AI without human authorship are not eligible for copyright protection (Thaler v. Perlmutter, 2023 confirms this direction). The practical question is what 'purely AI' means when a human writes a detailed prompt and edits the result. Third-party API terms add another layer: some providers grant users full ownership of outputs, some retain rights, and some require attribution. Organizations building products on AI APIs must review output ownership provisions specifically — building a commercial product on an API whose terms grant the provider rights to outputs creates significant legal exposure. These terms change; monitor them.

11. Patent Risk and Freedom-to-Operate Analysis

The AI patent landscape has grown rapidly: IBM, Google, Microsoft, Baidu, Samsung, and Qualcomm hold large AI patent portfolios covering neural network architectures, training techniques, inference optimizations, and application-specific implementations. Before deploying an AI system commercially, a freedom-to-operate (FTO) analysis — assessing whether the system would infringe any unexpired patent — is standard risk management for organizations with significant AI investments. Patent risk is asymmetric: the foundational techniques that matter most (transformer architectures, attention mechanisms, specific fine-tuning approaches) are exactly the ones most likely to have been patented by large technology companies. The SecAI+ exam tests awareness that patent risk exists and that FTO analysis is the appropriate organizational response, not the technical ability to conduct that analysis.

Key Terms Glossary

Term	Definition
Fair use	US copyright doctrine permitting certain uses of copyrighted material without license; four-factor test; not a guaranteed defense for AI training data.
Data provenance	Documentation of what data was used, where it came from, and what license governs its use in model training.
Trade secret	Confidential business information providing competitive advantage, protected by law as long as reasonable secrecy measures are maintained.
Model extraction attack	Querying a production model to collect input-output pairs used to train a substitute model that replicates the original's behavior.
Output watermarking	Embedding imperceptible signals in model outputs enabling forensic identification of models trained on those outputs.
Freedom-to-operate (FTO) analysis	Legal assessment of whether a system's operation would infringe unexpired patents held by third parties.
RAIL (Responsible AI License)	License framework adding behavioral restrictions on top of standard open-source terms, prohibiting specific harmful use cases.
Compartmentalization	Access control principle ensuring no single person has exposure to the complete IP picture — different individuals access different components.
Model memorization	A model's ability to reproduce training data (e.g., verbatim text or watermarked images) in its outputs; evidence of memorization strengthens copyright infringement claims.
ISO/IEC 42001:2023 §6.1	Requirement that IP risks be included in the AI management system's formal risk assessment process.

Quick Revision Checklist

- Training data copyright is actively litigated — do not assume fair use without legal analysis; memorization of training data strengthens plaintiff arguments.
- US fair use four factors: purpose (transformative?), nature (creative vs. factual?), amount taken, and market effect — factors 1 and 4 are dispositive in AI cases.
- Data provenance management (tracking what you trained on and under what license) is the operational risk control for training data copyright exposure.
- Model ownership requires explicit IP assignment clauses in employment, contractor, and research partnership agreements.
- Trade secrets (model weights, training data, methodology) require access controls, NDAs, compartmentalization, and audit trails to maintain protectable status.
- Model extraction attacks are a trade secret threat via API — defend with rate limiting, query anomaly detection, output watermarking, and prediction perturbation.
- Open-source AI licenses vary widely — Apache 2.0 is permissive; Meta LLaMA 2 and RAIL licenses carry behavioral restrictions that matter for commercial use.
- US Copyright Office position: purely AI-generated works without human authorship are not eligible for copyright protection.
- FTO analysis is required before commercial deployment of AI systems that use patented foundational techniques.
- ISO/IEC 42001:2023 Section 6.1 mandates IP risk as part of the AI management system's formal risk assessment.
- Third-party API output ownership terms and acceptable use policies must be reviewed by legal counsel before building commercial products.

10 Exam-Based MCQs — Intellectual Property Risks

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. ***Scenario:*** *Kamila is the AI governance lead at a media technology company preparing to train a large language model for content summarization. The engineering team proposes using a Common Crawl web scrape supplemented with books from Project Gutenberg and user-submitted feedback data. The CISO asks Kamila to assess the IP risk of each data source before training begins.* Which data source presents the HIGHEST intellectual property risk for commercial AI training?

A) Project Gutenberg texts, because they are in the public domain and widely reproduced.

B) Common Crawl web scrape, because it contains substantial copyrighted content without an explicit commercial training license.

C) User-submitted feedback data, because users may have submitted content that infringes third-party copyrights.

D) All three sources present equal IP risk because training data copyright law is fully settled.

Answer: B **Explanation:** B is correct because Common Crawl is a large-scale web scrape that includes substantial copyrighted content — news articles, books, academic papers — without any explicit license to use that content for commercial AI training. It is the primary focus of training data copyright litigation. A is wrong — Project Gutenberg hosts works in the public domain, making them safe for reproduction and training use. C is a real risk but secondary to the scale of copyrighted content in Common Crawl; user-submitted data is also typically covered by platform terms of service that grant rights to the platform. D is wrong — training data copyright law is actively contested and not settled as of 2025.

Q2. ***Scenario:*** *Darius is a security engineer at an AI startup that developed a proprietary recommendation model over 18 months. A competitor launches a nearly identical system three months after the startup's product goes public. Analysis of the competitor's outputs suggests their model was trained on the startup's API responses. The startup's model architecture and training data were never publicly disclosed.* Which MITRE ATLAS v2.1 technique BEST describes the attack used against the startup?

A) AML.T0020 — Poison Training Data
B) AML.T0036 — ML Model Inference API Access (model extraction)
C) AML.T0043 — Craft Adversarial Data
D) AML.T0049 — Prompt Injection

Answer: B **Explanation:** B is correct because the scenario describes systematic API querying to collect input-output pairs used to train a substitute model that replicates the proprietary model's behavior — the definition of a model extraction attack, classified under AML.T0036. A (Poison Training Data) involves corrupting training data, not extracting model behavior. C (Craft Adversarial Data) involves crafting inputs to cause specific model failures, not extraction. D (Prompt Injection) targets large language model instruction following, not model extraction.

Q3. ***Scenario:*** *Yuki is the general counsel at a technology company that uses the Meta LLaMA 2 model as the foundation for a commercial enterprise search product. The company has 12 million monthly active users. A developer on the team proposes using outputs from the LLaMA 2-based system to fine-tune a new foundation model the company is developing independently.* Which licensing restriction is MOST relevant to the developer's proposal?

A) The LLaMA 2 license prohibits commercial deployment for organizations with more than 700 million monthly active users.
B) The LLaMA 2 license prohibits using LLaMA 2 outputs to train competing foundation models.
C) The LLaMA 2 license requires all outputs to be attributed to Meta AI in the product interface.
D) The LLaMA 2 license is an Apache 2.0 license that permits all commercial and derivative uses.

Answer: B **Explanation:** B is correct because the Meta LLaMA 2 license explicitly prohibits using LLaMA 2 outputs to train competing foundation models — the developer's proposal to use LLaMA 2 outputs to fine-tune an independent foundation model would violate this restriction. A is wrong — the 700 million MAU threshold relates to whether commercial deployment itself is permitted, and the company with 12 million users is below that threshold; the specific concern here is the output-training restriction. C is incorrect — LLaMA 2 does not require attribution in the product interface. D is incorrect — LLaMA 2 uses a custom license, not Apache 2.0.

Q4. A company trains a generative AI image model on a dataset that includes watermarked stock photography. During testing, some generated images contain artifacts resembling stock photo watermarks. Which of the following represents the PRIMARY IP risk this finding indicates?

A) Patent infringement risk from using image generation techniques covered by third-party patents.

B) Trade secret misappropriation risk because competitor images were incorporated in training data.

C) Copyright infringement risk because model memorization of watermarked images is forensic evidence of unauthorized reproduction.

D) Output ownership risk because the generated images cannot be copyrighted due to AI authorship.

Answer: C **Explanation:** C is correct because the presence of watermarks in generated outputs is forensic evidence that the model memorized specific licensed images — exactly the type of evidence that strengthens training data copyright infringement claims, as seen in the Getty Images v. Stability AI litigation. Memorization weakens the transformative use / fair use defense significantly. A relates to patent risk, which is not raised by watermark artifacts. B describes trade secret misappropriation, which is not the risk indicated by watermark artifacts in outputs. D addresses output ownership, a real but separate concern from training data infringement.

Q5. Which of the following BEST describes the purpose of a freedom-to-operate (FTO) analysis in the context of AI system deployment?

A) Assessing whether the AI system's training data is licensed for commercial use.

B) Determining whether the AI system's operation would infringe any unexpired patents held by third parties.

C) Evaluating whether the AI system's outputs are eligible for copyright protection under current law.

D) Verifying that the AI system complies with applicable data privacy regulations in target jurisdictions.

Answer: B **Explanation:** B is correct because FTO analysis is a patent-specific assessment determining whether deploying and operating a system would infringe valid, unexpired patents — particularly relevant for AI given large portfolio holdings by IBM, Google, Microsoft, and others over foundational AI techniques. A describes data licensing due diligence, which is a separate activity addressing copyright risk rather than patent risk. C relates to output copyright eligibility under the US Copyright Office's human authorship requirements. D relates to privacy regulation compliance, not IP risk.

Q6. Which of the following actions is MOST effective at maintaining trade secret status for proprietary AI model weights?

A) Filing a patent application disclosing the model architecture and training methodology.

B) Publishing the model weights on an open-source platform under a restrictive license.

C) Implementing role-based access controls, NDAs, compartmentalization, and audit logs for model weight access.

D) Registering the model with the US Copyright Office as a creative work.

Answer: C **Explanation:** C is correct because trade secret protection requires 'reasonable efforts to maintain secrecy' — access controls, NDAs, compartmentalization, and audit logs collectively demonstrate that the organization took those efforts. Without these controls, trade secret protection may be lost even if the weights were never intentionally disclosed. A is the opposite of trade secret — filing a patent requires public disclosure of the protected invention, sacrificing trade secret protection. B publicly discloses the weights, destroying trade secret status regardless of the license terms. D is wrong — copyright registration relates to creative authorship, not confidential business information protection.

Q7. An organization developing an AI system engages a contractor to design the model architecture. After the project completes, the contractor claims ownership of the architecture they designed. Which contractual provision would have PREVENTED this dispute?

A) A non-compete clause prohibiting the contractor from working for competitors.

B) An explicit IP assignment clause in the contractor agreement assigning all work product IP to the organization.

C) A non-disclosure agreement preventing the contractor from disclosing the architecture to third parties.

D) A data processing agreement governing the contractor's use of personal data during development.

Answer: B **Explanation:** B is correct because absent an explicit IP assignment clause, the contractor may retain rights to work product they created, even when the organization paid for the work. An IP assignment clause in the contractor agreement specifically transfers ownership of created IP — including the model architecture — to the organization. A addresses competitive behavior after the engagement, not ownership of work product created during it. C prevents disclosure but does not transfer ownership — the contractor could retain rights and prohibit the organization from using the architecture without license. D addresses data privacy, not IP ownership.

Q8. Which of the following BEST describes the US Copyright Office's current position on copyright eligibility for AI-generated content?

A) AI-generated content is automatically protected by copyright owned by the organization that deployed the AI system.

B) AI-generated content is eligible for copyright protection if the AI model was trained on licensed data.

C) Works created purely by AI without human authorship are not eligible for copyright protection.

D) AI-generated content receives copyright protection for a reduced term of 14 years rather than the standard life-plus-70.

Answer: C **Explanation:** C is correct because the US Copyright Office has consistently held that copyright protection requires human authorship, and works created purely by AI are not eligible. This position was reinforced in Thaler v. Perlmutter (2023) and multiple subsequent Copyright Office registration rejections. A is wrong — the organization deploying the AI does not automatically receive copyright in purely AI-generated outputs. B is wrong — licensing of training data addresses training copyright risk, not output copyright eligibility. D is wrong — there is no reduced-term protection category for AI-generated works.

Q9. Which governance standard specifically requires that intellectual property risks be included in an AI management system's formal risk assessment?

A) NIST AI RMF 1.0 — GOVERN function

B) ISO/IEC 42001:2023 Section 6.1

C) EU AI Act (2024) Article 13

D) NIST CSF 2.0 — IDENTIFY function (ID.RA)

Answer: B **Explanation:** B is correct because ISO/IEC 42001:2023 Section 6.1 (Actions to address risks and opportunities) specifically requires that IP risks be identified and assessed as part of the AI management system's risk management process. This is the exam-tested citation for IP risk governance requirements. A (NIST AI RMF GOVERN) covers broader AI risk governance but is not the source of the specific IP risk assessment requirement. C (EU AI Act Article 13) covers transparency requirements for high-risk AI systems, not IP risk assessment. D (NIST CSF 2.0 ID.RA) addresses risk assessment broadly but is not the standard that specifically mandates IP risk inclusion.

Q10. An organization plans to defend its proprietary AI model from extraction attacks by embedding imperceptible signals in model outputs that persist in any model trained on those outputs. Which technique does this describe?

A) Query rate limiting

B) Input validation and sanitization

C) Model watermarking

D) Adversarial training

Answer: C **Explanation:** C is correct because output watermarking (also called model watermarking) embeds imperceptible signals in model outputs that are preserved when an adversary trains a substitute model on those outputs, enabling forensic identification of extracted models. This allows the IP owner to prove that a competitor's model was trained on their outputs in litigation or enforcement proceedings. A (query rate limiting) slows extraction by restricting queries per time period but does not provide forensic evidence of extraction. B (input validation) screens model inputs, not outputs. D (adversarial training) improves model robustness against adversarial perturbations, not model extraction.